

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 21: Trigonometrical Identities

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 The three basic identities

1. Assertion (A): If $\sin^2 A + \sin A = 1$, then $\cos^4 A + \cos^2 A = 1$. Reason (R): $1 - \sin^2 A = \cos^2 A$. (a) A true, R false (b) A false, R true (c) both true, and R is the correct reason for A (d) both true, but R is not the correct reason for A **ICSE 2025, 1 mark**

SOLUTION

From the given relation, $\sin A = 1 - \sin^2 A = \cos^2 A$ (using R).

Then $\cos^4 A + \cos^2 A = \sin^2 A + \sin A = 1$. So A is true, and R is exactly what proves it.

Answer: (c)

2. Statement 1: $\sin^2 \theta + \cos^2 \theta = 1$. Statement 2: $\operatorname{cosec}^2 \theta + \cot^2 \theta = 1$. Which is valid? (a) only 1 (b) only 2 (c) both (d) neither **ICSE 2024, 1 mark**

SOLUTION

Statement 1 is a basic identity. The correct form of the second is $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$, so statement 2 is not valid.

Answer: (a)

3. $(1 + \sin A)(1 - \sin A)$ equals: (a) $\operatorname{cosec}^2 A$ (b) $\sin^2 A$ (c) $\sec^2 A$ (d) $\cos^2 A$ **ICSE 2023, 1 mark**

SOLUTION

$(1 + \sin A)(1 - \sin A) = 1 - \sin^2 A = \cos^2 A$

Answer: (d)

2 Proving identities

4. Prove that $\frac{(\cot A + \tan A - 1)(\sin A + \cos A)}{\sin^3 A + \cos^3 A} = \sec A \operatorname{cosec} A$. **ICSE 2025, 4 marks**

SOLUTION

$$\begin{aligned}\cot A + \tan A - 1 &= \frac{\cos^2 A + \sin^2 A - \sin A \cos A}{\sin A \cos A} = \frac{1 - \sin A \cos A}{\sin A \cos A} \\ \sin^3 A + \cos^3 A &= (\sin A + \cos A)(\sin^2 A - \sin A \cos A + \cos^2 A) = (\sin A + \cos A)(1 - \sin A \cos A) \\ \text{LHS} &= \frac{(1 - \sin A \cos A)(\sin A + \cos A)}{\sin A \cos A (\sin A + \cos A)(1 - \sin A \cos A)} = \frac{1}{\sin A \cos A} \\ &= \sec A \operatorname{cosec} A = \text{RHS. Proved.}\end{aligned}$$

5. Factorise $\sin^3 \theta + \cos^3 \theta$. Hence prove that $\frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} + \sin \theta \cos \theta = 1$. **ICSE 2024, 4 marks**

SOLUTION

$$\begin{aligned}\text{Using } a^3 + b^3 &= (a + b)(a^2 - ab + b^2): \sin^3 \theta + \cos^3 \theta = (\sin \theta + \cos \theta)(\sin^2 \theta - \sin \theta \cos \theta + \cos^2 \theta) \\ &= (\sin \theta + \cos \theta)(1 - \sin \theta \cos \theta) \\ \text{LHS} &= \frac{(\sin \theta + \cos \theta)(1 - \sin \theta \cos \theta)}{\sin \theta + \cos \theta} + \sin \theta \cos \theta = 1 - \sin \theta \cos \theta + \sin \theta \cos \theta \\ &= 1 = \text{RHS. Proved.}\end{aligned}$$

6. Prove that $(\sin^2 \theta - 1)(\tan^2 \theta + 1) + 1 = 0$. **ICSE 2023, 4 marks**

SOLUTION

$$\begin{aligned}\sin^2 \theta - 1 &= -\cos^2 \theta \text{ and } \tan^2 \theta + 1 = \sec^2 \theta \\ \text{LHS} &= -\cos^2 \theta \times \sec^2 \theta + 1 = -1 + 1 \\ &= 0 = \text{RHS. Proved.}\end{aligned}$$

7. Prove that $\frac{\sin A}{1 + \cot A} - \frac{\cos A}{1 + \tan A} = \sin A - \cos A$. **ICSE 2020, 3 marks**

SOLUTION

$$\begin{aligned}\frac{\sin A}{1 + \cot A} &= \frac{\sin A}{\frac{\sin A + \cos A}{\sin A}} = \frac{\sin^2 A}{\sin A + \cos A} \text{ and } \frac{\cos A}{1 + \tan A} = \frac{\cos^2 A}{\sin A + \cos A} \\ \text{LHS} &= \frac{\sin^2 A - \cos^2 A}{\sin A + \cos A} = \frac{(\sin A - \cos A)(\sin A + \cos A)}{\sin A + \cos A} \\ &= \sin A - \cos A = \text{RHS. Proved.}\end{aligned}$$

8. Prove that $\left(\frac{1 - \tan \theta}{1 - \cot \theta}\right)^2 = \tan^2 \theta$. **ICSE 2020, 4 marks**

SOLUTION

$$1 - \cot \theta = 1 - \frac{1}{\tan \theta} = \frac{\tan \theta - 1}{\tan \theta}$$

$$\frac{1 - \tan \theta}{1 - \cot \theta} = \frac{(1 - \tan \theta) \tan \theta}{\tan \theta - 1} = -\tan \theta$$

Squaring: LHS = $(-\tan \theta)^2 = \tan^2 \theta = \text{RHS}$. **Proved.**

9. Prove that $(\operatorname{cosec} \theta - \sin \theta)(\sec \theta - \cos \theta)(\tan \theta + \cot \theta) = 1$. **ICSE 2019, 3 marks**

SOLUTION

$$\operatorname{cosec} \theta - \sin \theta = \frac{1 - \sin^2 \theta}{\sin \theta} = \frac{\cos^2 \theta}{\sin \theta} \text{ and } \sec \theta - \cos \theta = \frac{\sin^2 \theta}{\cos \theta}$$

$$\tan \theta + \cot \theta = \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} = \frac{1}{\sin \theta \cos \theta}$$

$$\text{LHS} = \frac{\cos^2 \theta}{\sin \theta} \times \frac{\sin^2 \theta}{\cos \theta} \times \frac{1}{\sin \theta \cos \theta} = 1 = \text{RHS. Proved.}$$

10. Prove that $\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta} = \tan \theta + \cot \theta$. **ICSE 2018, 3 marks**

SOLUTION

$$\sec^2 \theta + \operatorname{cosec}^2 \theta = \frac{1}{\cos^2 \theta} + \frac{1}{\sin^2 \theta} = \frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta \cos^2 \theta} = \frac{1}{\sin^2 \theta \cos^2 \theta}$$

$$\text{LHS} = \frac{1}{\sin \theta \cos \theta}$$

$$\text{RHS} = \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} = \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} = \frac{1}{\sin \theta \cos \theta}$$

LHS = RHS. **Proved.**

11. Prove that $(1 + \cot \theta - \operatorname{cosec} \theta)(1 + \tan \theta + \sec \theta) = 2$. **ICSE 2018, 4 marks**

SOLUTION

$$\begin{aligned} \text{LHS} &= \left(\frac{\sin \theta + \cos \theta - 1}{\sin \theta}\right) \left(\frac{\cos \theta + \sin \theta + 1}{\cos \theta}\right) = \frac{(\sin \theta + \cos \theta)^2 - 1}{\sin \theta \cos \theta} \\ &= \frac{1 + 2 \sin \theta \cos \theta - 1}{\sin \theta \cos \theta} = 2 = \text{RHS. Proved.} \end{aligned}$$

12. Prove that $\frac{\sin \theta - 2 \sin^3 \theta}{2 \cos^3 \theta - \cos \theta} = \tan \theta$. **ICSE 2017, 3 marks**

SOLUTION

$$\text{LHS} = \frac{\sin \theta (1 - 2 \sin^2 \theta)}{\cos \theta (2 \cos^2 \theta - 1)}$$

Using $1 = \sin^2 \theta + \cos^2 \theta$: $1 - 2 \sin^2 \theta = \cos^2 \theta - \sin^2 \theta$ and $2 \cos^2 \theta - 1 = \cos^2 \theta - \sin^2 \theta$.

The brackets cancel: LHS = $\frac{\sin \theta}{\cos \theta} = \tan \theta = \text{RHS}$. **Proved.**

3 Complementary angles

- 13.** Without using tables, evaluate $\sin^2 28^\circ + \sin^2 62^\circ + \tan^2 38^\circ - \cot^2 52^\circ + \frac{1}{4} \sec^2 30^\circ$. **ICSE 2017, 3 marks**

SOLUTION

$\sin 62^\circ = \cos 28^\circ$, so $\sin^2 28^\circ + \sin^2 62^\circ = \sin^2 28^\circ + \cos^2 28^\circ = 1$.

$\cot 52^\circ = \tan 38^\circ$, so $\tan^2 38^\circ - \cot^2 52^\circ = 0$.

$$\frac{1}{4} \sec^2 30^\circ = \frac{1}{4} \times \frac{4}{3} = \frac{1}{3}$$

Answer: $1 + 0 + \frac{1}{3} = \frac{4}{3}$

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